

SIMATS ENGINEERING

ASSESSMENT TOOL 1

Experiments

COURSE NAME: Computer Networks

COURSE CODE: CSA0702

COURSE OUTCOME COVERED

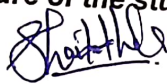
CO3: Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions

Assessment Tool Weightage: 40%

Code of Conduct:

I SHAIK KHATAH ROSHAN 192511445 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student



QUESTIONS:4

Experiments

Total Marks:40

1. Capture packets during a TCP connection setup. Identify the three packets involved in the handshake (SYN, SYN-ACK, ACK). Demonstrate the role of each flag and how they establish a reliable connection. (BL4) (10)
2. Capture a DNS query using UDP in Wireshark. Compare the UDP header size with TCP and note the absence of sequence numbers or acknowledgments. Discuss how this lightweight design impacts speed and reliability. (BL4) (10)
3. Simulate packet loss (e.g., disconnect briefly) and capture traffic in Wireshark. Observe how TCP retransmits lost packets while UDP does not. Explain why this difference makes TCP reliable but slower compared to UDP. (BL4) (10)
4. Use Wireshark to capture DNS traffic while resolving a domain name. Identify the query type (A, AAAA, MX) and the response received. Measure the time taken for resolution and explain why DNS typically uses UDP.(BL4) (10)

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

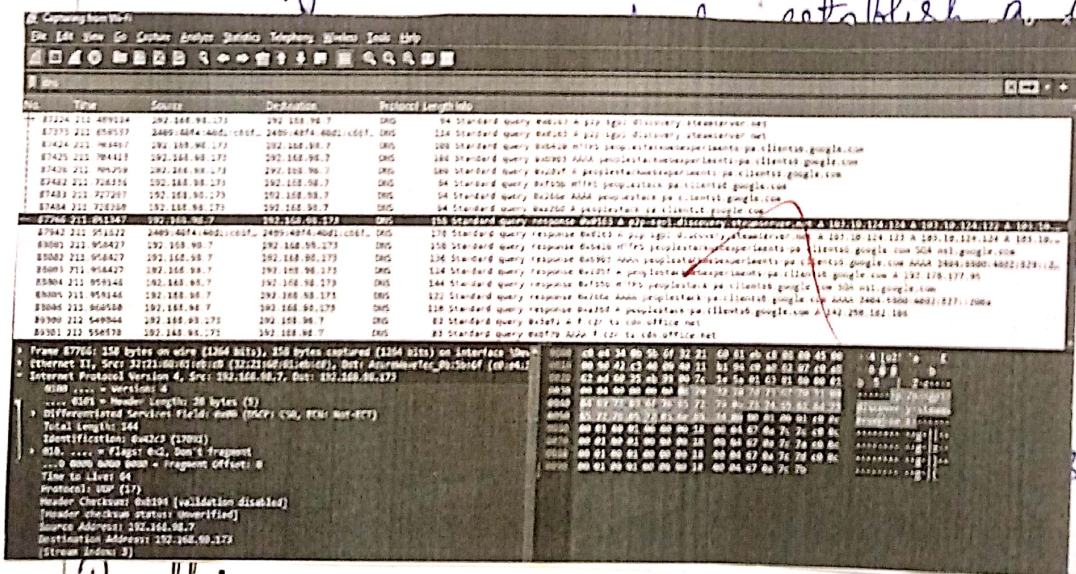
Q1. TCP Three-Way Handshake

Aim:

To capture packets during TCP connection setup using Wireshark and identify the SYN, SYN-ACK, and ACK packets involved in establishing a reliable connection.

Procedure:

1. Open Wireshark and select the active network interface.
2. Start packet capture.
3. Open a web browser and connect to a website.
4. Apply the filter: tcp
5. Identify the three packets:



Result:

The TCP three-way handshake was successfully captured. The SYN, SYN-ACK, and ACK packets were identified.

demonstrating how TCP establishes a reliable connection before data transmission.

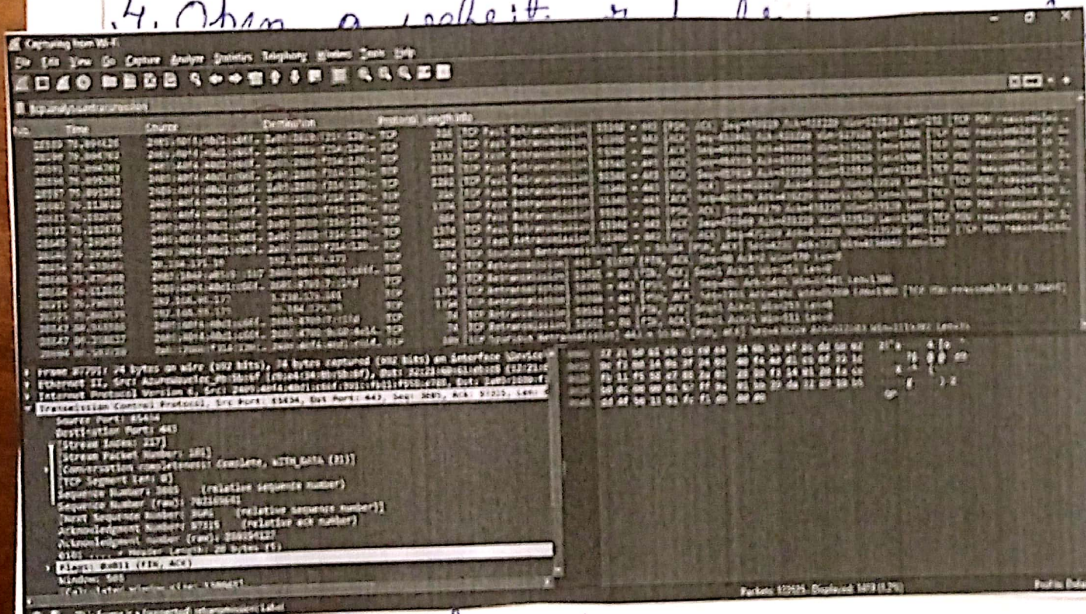
Q2. DNS Query Using UDP

Aim:-

To capture a DNS query using UDP in Wireshark and compare the UDP header with the TCP header to understand why UDP is faster but less reliable.

Procedure:

1. Open Wireshark and select the active network interface.
2. Start packet capture.
3. Apply the filter: dns
4. Open a web browser and visit a website.



9. Record the observations.

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Result:

The DNS query was successfully captured using UDP, UDP has a smaller and simpler header than TCP and does not use sequence numbers or acknowledgments. Therefore, UDP provides faster communication with lower overhead, but it does not provide TCP-level reliability.

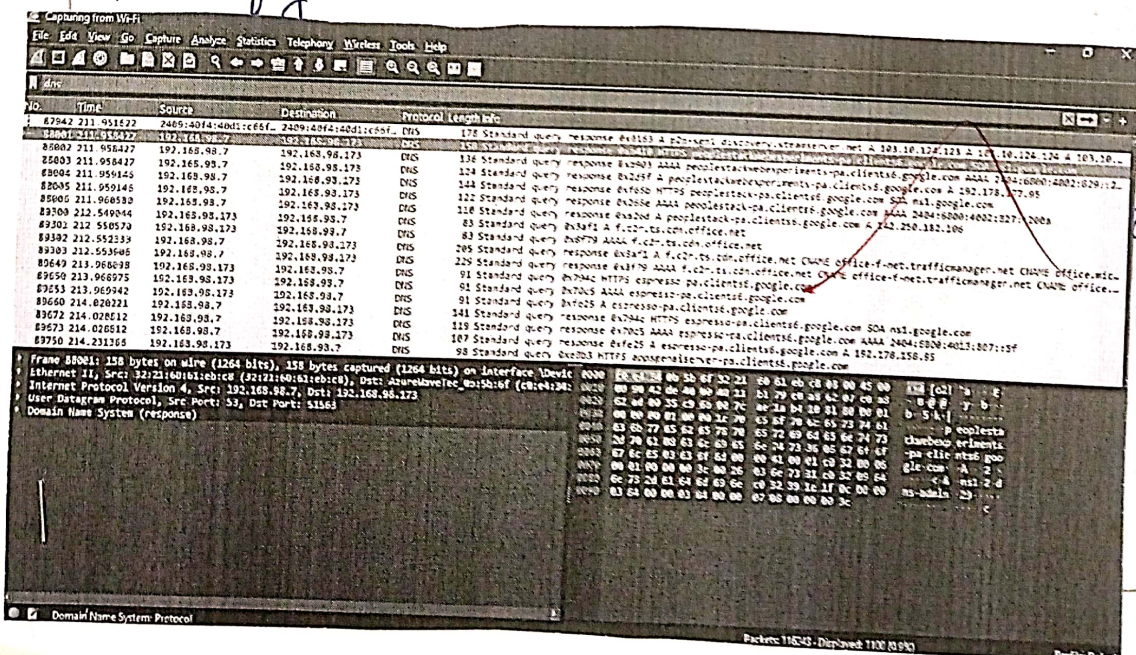
Q 3, TCP Retransmission vs UDP

Aim:

To simulate packet loss and observe how TCP retransmits lost packets while UDP does not.

Procedure:

- 1, Open Wireshark and start packet capture.
- 2, Begin a TCP or UDP communication.
- 3, Briefly disconnect the network connection to



resumes.

9, Note whether UDP retransmits the lost packets.

Result:

TCP retransmitted packets that were lost during transmission, whereas UDP did not retransmit them. This shows that TCP provides reliable delivery, but its retransmission and acknowledgment mechanisms introduce additional delay. UDP avoids this overhead and is therefore generally faster but less reliable.

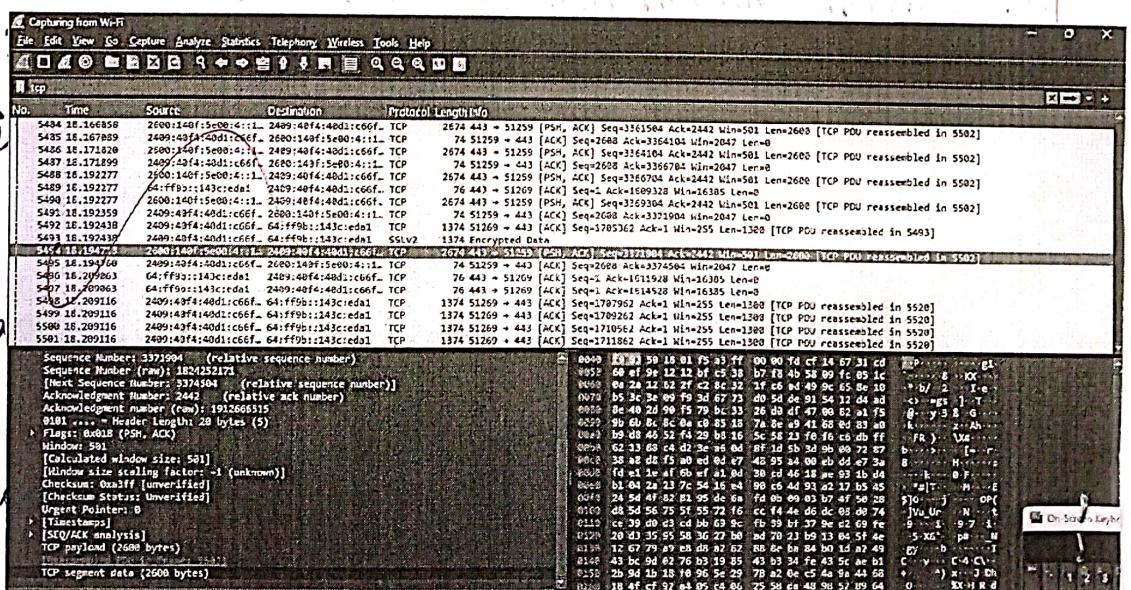
Q4. DNS Traffic and Resolution Time.

Aim:

To capture DNS traffic using Wireshark, identify the DNS query type and response, measure the resolution time, and understand why DNS commonly uses UDP.

Procedure

- 1, Open Wireshark
- 2, Start packet capture on the network interface.
- 3, Open a web browser and visit a website that has not been visited recently.
- 4, Apply the filter 'dns' to capture only DNS traffic.



5, Identify the DNS query packet.

6, Check the Query Type. Such as:

- A - IPv4 address

- AAAA - IPv6 address

- MX - Mail exchange server

7, Identify the corresponding DNS response.

8, Record the requested domain and returned address / record.

9, Measure the time between the DNS query and its response.

10, Stop the capture and analyze the packets.

Result:

The DNS query and corresponding responses were successfully captured. The query type (A, AAAA, or MX) and response were identified, and the DNS resolution time was measured. DNS commonly uses UDP because its lightweight design provides low overhead and fast query - response communication.

Q.2.2